RF-Free Quantum Sensing via Strain-Induced Spin Control in NV Centers

Luke Simunic*
Professor Andrew D. Greentree†
Qiang Sun‡

May 2025

Abstract

This project investigates RF-free quantum control in nitrogen-vacancy (NV) centers in diamond through time-varying strain, simulating phonon-driven spin transitions using photothermal excitation. A spin-1 Hamiltonian was constructed to incorporate zero-field splitting, Zeeman effect, and transverse strain coupling. A time-dependent Schrodinger equation was numerically solved to simulate the systems eigenvalue evolution and population dynamics under oscillatory strain. Results revealed that dynamic strain at resonant frequencies (∼500 MHz) can drive coherent, Rabi-like oscillations between spin sublevels, replicating the key features of conventional ODMR without the need for radiofrequency fields. The strain-induced transitions were shown to be sharply frequency-selective and coherent over extended timescales. These findings demonstrate the feasibility of phonon-driven spin control and offer a foundation for future strain-based quantum sensing applications in environments where RF delivery is constrained.

Background and Introduction

Nitrogen-vacancy (NV) centers in diamond are quantum systems with desirable spin properties that make them valuable for applications in quantum sensing, magnetometry, and quantum information processing. There is currently a major interest in the application of NV centers for Optically Detected Magnetic Resonance (ODMR). ODMR produces a spectrum which enables precise measurement of magnetic fields, electric fields, temperature, and strain at the nanoscale. The NV center is an atomic defect within the lattice structure of a diamond, its unique structure

*Undergraduate Researcher, RMIT University
†School of Science, RMIT University
‡Research Fellow, RMIT University

exhibits desirable quantum properties for performing ODMR spectroscopy. The NV^- charge state consists of a spin pair ($m_s = \pm 1$) that exhibits long coherence times at room temperature. It is a confined electronic state with a triplet spin ($S = 1$) and is the only currently known charge state shown to be ODMR active.

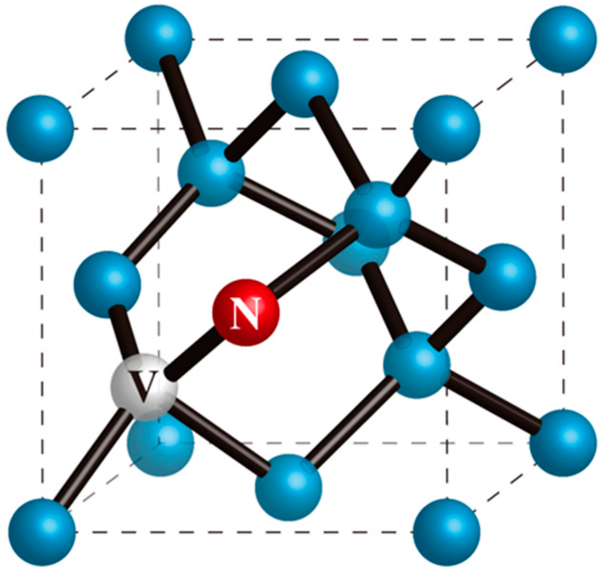

Figure 1: *Schematic of an NV center within a diamond lattice. Carbon atoms are in blue, the Nitrogen atom is in red and is adjacent to the vacancy within the lattice in grey. Figure reproduced from (Acosta, 2014)*

ODMR in the negatively charged nitrogen-vacancy (NV^-) center in diamond exploits spin-dependent fluorescence to enable quantum sensing at room temperature. The NV^- center exhibits a zero-field splitting $D \approx 2.87$ GHz between the $m_s = 0$ and $m_s = \pm 1$ sublevels due to spin-spin dipolar interactions (Doherty et al., 2013). The system is first optically initialized using a 532 nm green laser, which polarizes the spin into the $m_s = 0$ state via a non-radiative intersystem crossing through a metastable singlet pathway (Jelezko and Wrachtrup, 2006). A swept microwave (MW) or radio (RF) field is then applied to induce transitions between $m_s = 0$ and $m_s = \pm 1$ states. At resonance (i.e., when the MW/RF frequency matches the spin transition frequency), the population is transferred out of the $m_s = 0$ state, leading to a measurable reduction in fluorescence intensity.

Introducing the presence of an external magnetic field B_z, Zeeman splitting causes the degeneracy of the $m_s = \pm 1$ levels to lift, producing two distinct resonance dips centered around $D \pm \gamma_e B_z$, where γ_e is the electron gyromagnetic ratio. The Zeeman effect describes the lifting of degeneracy in magnetic sublevels due to an external magnetic field, a phenomenon arising from the interaction of the magnetic moment associated with spin angular momentum and the applied field. In the NV^- center's ground-state spin triplet, the application of a magnetic field

$\vec{B}$ introduces an additional Hamiltonian term:

$$H_{\text{Zeeman}} = \gamma_e \vec{B} \cdot \vec{S} \tag{1}$$

Where γ_e is the electron gyromagnetic ratio and $\vec{S}$ is the spin-1 operator. For a field aligned along the NV axis (z-axis), this leads to a linear shift of the $m_s = \pm 1$ levels according to $E = D \pm \gamma_e B_z$, while the $m_s = 0$ state remains unaffected to first order.

In addition to zero-field and Zeeman splitting, hyperfine interactions in the NV$^-$ center arise from magnetic dipole coupling between the electronic spin ($S = 1$) and proximal nuclear spins, most notably the host ^{14}N nucleus ($I = 1$) and nearby ^{13}C nuclei ($I = 1/2$). These interactions introduce additional terms into the Hamiltonian of the form:

$$H_{\text{HF}} = \vec{S} \cdot \mathbf{A} \cdot \vec{I} \tag{2}$$

Where $\mathbf{A}$ is the hyperfine interaction tensor, often assumed to be axial for the on-site ^{14}N nucleus. This coupling lifts the degeneracy of the electronic spin states by further splitting each m_s sublevel into hyperfine-resolved components depending on the nuclear spin projection m_I. Consequently, the energy eigenstructure becomes a dense manifold of hyperfine levels, which must be considered in high-resolution spectroscopy and quantum control. In ODMR experiments, these splittings manifest as multiple, closely spaced resonances superimposed on the main electron spin transitions, enabling nuclear-spin-resolved readout and coherent manipulation protocols (Doherty et al., 2013; Smeltzer et al., 2009).

Furthermore, strain-induced shifts that arise due to lattice distortion in the diamond crystal further modify the electronic energy levels and spin transition frequencies of the NV center. Strain in the diamond lattice perturbs the local crystal field symmetry at the NV$^-$ center site, resulting in mixing and splitting of the $m_s = \pm 1$ sublevels even in the absence of an external magnetic field. This effect is incorporated into the spin Hamiltonian via the transverse strain term:

$$H_{\text{strain}} = E(S_x^2 - S_y^2) \tag{3}$$

Where E is the effective strain coupling coefficient and S_x, S_y are spin-1 operators transverse to the NV axis. The origin of this strain can be intrinsic, arising from crystal lattice imperfections,

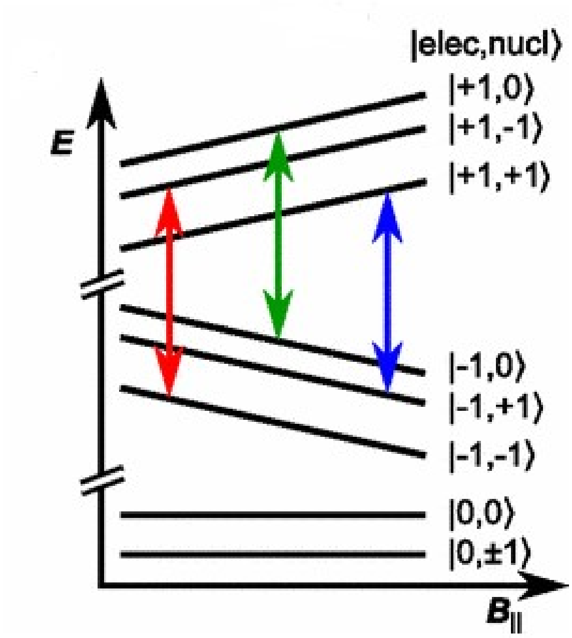

Figure 2: *The hyperfine structure of the NV center with labels indicating transitions between states. The inclusion of the nuclear spin moves the entire system from 3 dimensional to 9. Figure reproduced from (Acosta, 2014)*

vacancy clustering, or proximity to grain boundaries in polycrystalline diamond. More interestingly, strain can be introduced extrinsically, due to externally applied stress, nanofabrication-induced defects, or thermal expansion. This anisotropic distortion breaks the axial symmetry of the zero-field splitting, lifting the degeneracy of the $|m_s = \pm 1\rangle$ states and modifying the eigenvalue structure of the system. In ODMR, this manifests as a splitting of the ± 1 resonance lines at zero magnetic field, while dynamic modulation of the strain, via mechanical or photothermal excitation, offers a route to coherently access forbidden transitions and engineer tailored spin-phonon interactions.

This mechanical driving of spin transitions in the NV^- center exploits time-varying lattice strain to induce coherent modulation of the spin Hamiltonian. In this context, externally applied resonant mechanical vibrations, typically at gigahertz frequencies, generate phonons that couple to the spin states through strain-induced modifications of the crystal lattice. The interaction is captured by introducing a time-dependent strain term to the Hamiltonian:

$$H_{\mathrm{strain}}(t) = E(t)(S_x^2 - S_y^2) \tag{4}$$

Where $E(t) = E_0 \cos(\omega t)$ represents an oscillatory transverse strain field. When the modulation frequency ω matches the energy splitting between spin sublevels, this strain-mediated coupling can drive transitions between $m_s = 0$ and $m_s = \pm 1$ states. Crucially, because the strain field transforms as a second-rank tensor, it enables access to otherwise magnetically forbidden transitions (between $|m_s = +1\rangle$ and $|m_s = -1\rangle$), that violate magnetic dipole selection rules. This provides a powerful alternative to microwave or radio frequency magnetic fields in spin manipulation, especially in contexts where such systems are intrusive and costly, such as cryogenic, biological, or electrically isolated environments (MacQuarrie et al., 2015; Barfuss et al., 2015; Teissier et al., 2014a).

In the spin Hamiltonian of the (NV^-) center, mechanical strain introduces additional off-diagonal terms that modify the energy levels of the ground-state spin triplet. Strain effects are typically represented in terms of a perturbation to the zero-field splitting D_0, and are expressed using effective parameters $\sigma_\parallel$, σ_x, and σ_y, all in units of frequency (Hz). These terms originate from coupling between the NV center's spin and lattice deformations, where $\sigma_\parallel$ modifies the diagonal energies, reflecting axial (longitudinal) strain aligned with the NV axis, and σ_x, σ_y contributes to the off-diagonal components of the Hamiltonian, encoding transverse strain-induced mixing of the spin states. At a microscopic level, this coupling arises from the interaction between the spin operators S_iS_j and components of the strain tensor ϵ_{ij} through second-rank coupling constants d_{ij}, which describe the NV centers strain susceptibility. These strain terms are typically related to mechanical stress σ_{ij} via Hookes law, $\sigma_{ij} = C_{ijkl}\epsilon_{kl}$, where C_{ijkl} is the diamond elastic stiffness tensor. Therefore, while the Hamiltonian treats strain terms as distinct constants, they are derived from mechanical deformation and stress in the crystal lattice, whether intrinsic (e.g., lattice mismatch or defects) or externally driven (e.g., via acoustic waves or photothermal expansion) (Acosta, 2014).

Recent advances have demonstrated that time-varying mechanical strain, through resonant phonon driving, can induce coherent control of NV spin states, enabling transitions that are otherwise magnetically forbidden, such as $\Delta m_s = \pm 2$ (Ovartchaiyapong et al., 2013; Maity et al., 2024). These phonon-driven transitions exploit the direct spin-phonon coupling enabled by strain modulation, and open novel pathways for all-mechanical quantum control protocols and hybrid quantum devices.

While optically detected magnetic resonance (ODMR) underpins the operation of NV-based quantum magnetometers by enabling spin-state readout through radiofrequency (RF) or microwave-induced transitions, it is fundamentally reliant on the delivery of high-frequency electromagnetic fields to the quantum system. This dependence imposes limitations in scenarios where RF penetration is poor, such as in conductive or high-permittivity media (e.g., biological tissues, metals, or underwater environments), or where field delivery is spatially constrained. Moreover, the generation and control of RF fields require additional hardware, increasing system complexity, energy consumption, and susceptibility to electromagnetic interference.

These constraints motivate the search for alternative spin control mechanisms that do not rely on radiofrequency fields. In particular, applications in biomedical imaging, in vivo sensing, or underwater magnetometry demand minimally invasive, compact, and RF-free platforms. By eliminating the need for direct RF excitation, we could simplify hardware requirements, reduce power consumption, and enable operation in environments where RF-based systems would otherwise fail or introduce noise. Exploring mechanical or photothermal pathways for spin manipulation thus opens promising avenues for next-generation quantum sensing technologies.

The use of mechanical strain waves, phonons, is one of these alternative to drive spin transitions through direct spin-phonon coupling (Golter et al., 2016; MacQuarrie et al., 2013). This approach has the potential to bypass the need for RF fields by leveraging strain-induced modulation of the crystal field environment to drive transitions between NV spin sublevels, including those that are otherwise magnetically forbidden. However, this area lacks research, particularly regarding quantitative validation of the effectiveness of phonon driving in reliably achieving coherent control. Furthermore, the parameter regimes (strain amplitude, mechanical resonance matching, and temporal coherence) required to induce sufficiently strong coupling for practical magnetometry are not yet well characterized (Barfuss et al., 2024). This project addresses this gap by simulating time-dependent modulation of the NV Hamiltonian under phonon-like strain to assess whether phonon-driven transitions can achieve energy-level splittings analogous to those traditionally induced by RF excitation.

This project explores whether photothermal excitation can dynamically modulate the crystal field splitting in an NV center, thereby controlling transitions into spin states. The ability to reach these states could expand the scope of NV-based quantum technologies, offering new pathways for spin control and quantum- enhanced sensing. We aim to quantify the extent to which phonon-induced modulations enable otherwise forbidden transitions. The results will provide insight into the feasibility of phonon- driven spin manipulation and its potential applications in solid-state quantum systems.

Aims

The aim of this project is to measure the effect of phonon driving using photo-thermal excitation to drive spin transitions within a diamond nitrogen-vacancy (NV) center. To evaluate the feasibility of RF-free spin control, a computational simulation was developed to solve the time-dependent Schrodinger equation (TDSE) for an NV^- center under dynamic strain. This approach investigates whether photothermal excitation can drive transitions and coherently modulate the NV spin Hamiltonian to enable transitions between spin sublevels. The study further aims to quantify the conditions under which such phonon-induced spin control becomes comparable to, or exceeds, the effectiveness of conventional RF-driven transitions.

The objectives of this project are:

- Computationally simulate the time-dependent Schrodinger equation (TDSE) to observe the NV centers spin states evolve over time
- Systematically vary key parameters to identify resonant conditions where spin state transitions are achievable
- Measure the effect of an induced strain on the Nitrogen Vacancy center's energy levels
- Measure the population dynamics of the three state system under an oscillating strain effect
- Measure the magnitude of the population transitions across different strain frequencies
- Visualize the results as energy level diagrams, energy levels over time and population transfer models to assess the feasibility of phonon-driven transitions

Methodology

To evaluate the feasibility of RF-free spin control via photothermal excitation, a computational simulation was developed to solve the time-dependent Schrodinger equation (TDSE) for an NV^- center in vacuum.

Modeling the Hamiltonian

The NV center's ground state is a spin-triplet ($S = 1$) system. The NV center's spin Hamiltonian in the presence of magnetic fields and strain is given by:

$$H(t) = \left(D_0 + \epsilon_\parallel \sigma_\parallel(t)\right) S_z^2 + \gamma_{\mathrm{NV}} B_\parallel S_z + \gamma_{\mathrm{NV}} B_\perp S_x - \epsilon_\perp \sigma_x(t)(S_x^2 - S_y^2) + \epsilon_\perp \sigma_y(t)(S_x S_y + S_y S_x)$$

Where:

- $D_0 = 2.87\,\mathrm{GHz}$ is the zero-field splitting constant

- $\gamma_{\rm NV} = 2.8\,{\rm MHz\,G^{-1}}$ is the NV center gyromagnetic ratio
- $B_{\parallel}$, $B_{\perp}$ are the magnetic field components
- $\epsilon_{\perp} = 0.03\,{\rm MHz\,MPa^{-1}}$ is the perpendicular strain coupling constant
- $\sigma_x(t)$ and $\sigma_y(t)$ are time-dependent strain modulations induced by photothermal excitation
- S_x, S_y, S_z are spin-1 operators

This equation was based on the Hamiltonian derived for the mechanical spin control of an NV center (MacQuarrie et al. (2013)). The derivation of this Hamiltonian can be found in the Appendix A.

We incorporated the Hamiltonian in matrix form within the python simulation as:

$$H_{NV} = \begin{bmatrix} (D_0 + \epsilon_{\parallel}\sigma_{\parallel} + \gamma_{NV}B_{\parallel}) & (\frac{\gamma_{NV}B_{\perp}}{\sqrt{2}} - i\epsilon_{\perp}\sigma_y) & (\epsilon_{\perp}\sigma_x) \\ (\frac{\gamma_{NV}B_{\perp}}{\sqrt{2}} + i\epsilon_{\perp}\sigma_y) & 0 & (\frac{\gamma_{NV}B_{\perp}}{\sqrt{2}} - i\epsilon_{\perp}\sigma_y) \\ (\epsilon_{\perp}\sigma_x) & (\frac{\gamma_{NV}B_{\perp}}{\sqrt{2}} + i\epsilon_{\perp}\sigma_y) & (D_0 + \epsilon_{\parallel}\sigma_{\parallel} - \gamma_{NV}B_{\parallel}) \end{bmatrix}$$

To validate strain-induced effects, we computed the instantaneous eigenvalues of H_{NV} to visualize splitting, level crossings, anti-crossings, and adiabatic versus non-adiabatic transitions. This allowed us to track how the strain perturbs the spin levels.

In the absence of external RF fields, $\sigma_x(t)$ and $\sigma_y(t)$ serve as the driving mechanisms for spin transitions via lattice vibrations, emulating coherent phonon interaction. We introduced the hyperfine strain as a constant (A_{iso}) to keep the Hamiltonian in a 3 dimensional Hermitian space. Introduction of the Hyperfine splitting required incorporating the six nuclear spin states, moving the Hamiltonian from 3 dimensional to 9 dimensional Hermitian space. This resulted in the simulation becoming unstable and remains a limitation that could be further developed. As a result, we assumed a constant hyperfine strain which proved sufficient.

Photothermal strain was modeled as a sinusoidal driving field:

$$\sigma_x(t) = A\cos(\omega t), \quad \sigma_y(t) = A\sin(\omega t)$$

This circular modulation emulated coherent lattice distortion, driving shifts in the NVs spin sublevels through off-diagonal terms.

We explored the following variable ranges:

- Strain frequency (ω): 0.1 MHz to 9 GHz

- Strain amplitude (σ_x, σ_y): 1×10^5 Hz to 1×10^9 Hz
- Magnetic field orientation: $B_{\parallel}$ and $B_{\perp}$ from 0 to 100 G

We simulated and plotted the energy levels of each eigenvalue against a parallel magnetic field from a range of $-100G$ to $100G$. A perpendicular magnetic field was introduced and the strain was varied until an anti-crossing was observed.

We then incorporated the time-dependent strain to act as the photothermal strain, as an additional term:

$$\epsilon(t)(S_x^2 - S_y^2)$$

with

$$\epsilon(t) = A\cos(\omega t)$$

where A is the strain amplitude and ω is the driving frequency, matching the energy gap between spin sublevels.

The simulation was then used to model the eigenvalue ($m_s = 0$ and $m_s = \pm1$) energy levels as a function of time. All perpendicular elements were set to zero. An oscillating strain (σ_x,σ_y) and a parallel magnetic field ($B_{\parallel}$) were introduced. The parallel magnetic field was kept within the range of the anti-crossing determined by the prior graphs ($\approx -10G$ to $\approx 10G$)

TDSE Simulation

The time-dependent Schrodinger equation (TDSE) governs the evolution:

$$i\hbar\frac{d}{dt}\left|\psi(t)\right\rangle = H(t)\left|\psi(t)\right\rangle$$

The system was initialized in the $m_s = 0$ state and evolved using a Runge-Kutta method, specifically RK45, to ensure accurate time-stepping over nanosecond to microsecond time scales. Population dynamics were tracked by computing $|\langle m_s|\psi(t)\rangle|^2$ for each sub-level. The RK45 method offers adaptive step sizing which balances computational efficiency with precision, particularly in regions of rapid oscillations. However, since RK45 is optimized for smooth problems (continuous and well defined), it can become inefficient when dealing with highly stiff systems, discontinuous modulations in the Hamiltonian or when high-frequency strain drives induce fast oscillations in the wavefunction. As a result the system requires additional safeguards or step-size constraints to maintain numerical stability. This pertains to the optimization limitations of the system that could be improved in the future.

Numerical Solution

To simulate the quantum dynamics of the NV center under photothermal strain, the time-dependent Schrodinger equation (TDSE) was solved numerically using a custom Python implementation (see Listing 1 in Appendix B). The NV center Hamiltonian is defined in the spin-1 basis $\{|+1\rangle, |0\rangle, |-1\rangle\}$, incorporating magnetic field components and time-dependent strain-induced terms.

The initial spin state $|\psi(0)\rangle$ is prepared by projecting a chosen population distribution onto the instantaneous eigenbasis at $t = 0$. The TDSE:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

is integrated using the 'RK45' solver from SciPy's 'solve_ivp' library, with tight tolerances (`rtol` = `atol` = 10^{-8}) to maintain high numerical accuracy. RK45 provides adaptive step sizing, which is well-suited to capturing both slow and rapid spin dynamics over microsecond-scale simulation windows.

At each timestep, the instantaneous Hamiltonian $H(t)$ is diagonalized to compute the time-evolving eigenbasis. The evolving state $|\psi(t)\rangle$ is then projected onto the eigenvectors to calculate the population in each eigenstate:

$$P_j(t) = | \langle \phi_j(t) | \psi(t) \rangle |^2$$

Where $|\phi_j(t)\rangle$ denotes the j-th eigenvector of $H(t)$ at time t.

This method enabled direct visualisation of spin transitions and allowed identification of resonant driving conditions. We swept the driving frequency until we observed conditions where population transitioned from the $\langle m_s = 0\rangle$ to the $\langle m_s = \pm 1\rangle$ states. We continued until we observed significant driving of spin transitions corresponding to resonance.

Simulation Outputs

The simulation computed static energy level evolution across a range of magnetic field values, energy level evolution under dynamic strain (via instantaneous Hamiltonian eigenvalues), population dynamics of the spin states $|m_s = 0\rangle, |m_s = \pm 1\rangle$, and the transition probabilities as a function of time and frequency

The simulation was performed using Python 3.10 with `numpy`, `scipy`, and `matplotlib`, Pyto iOS app and Pycharm Community Edition for programming and visualisation, and Git for version control of simulation parameters and test cases. All simulations were run on a standard CPU architecture without GPU acceleration due to the small Hilbert space of the spin-1 system.

Results and Analysis

The energy level structure of the NV center in diamond was simulated by solving the eigenstates of the Hamiltonian (H_{NV}). Figures 3 and 4 display the energy of the spin sublevels $m_s = 0, \pm 1$ calculated from the NV center Hamiltonian, incorporating zero-field splitting, Zeeman splitting and strain-induced coupling effects. The energy level of each was plotted as a function of an external, parallel magnetic field ($B_{\|}$). Three varying magnitudes of strain were applied to display the splitting effect of strain on the energy levels of the system.

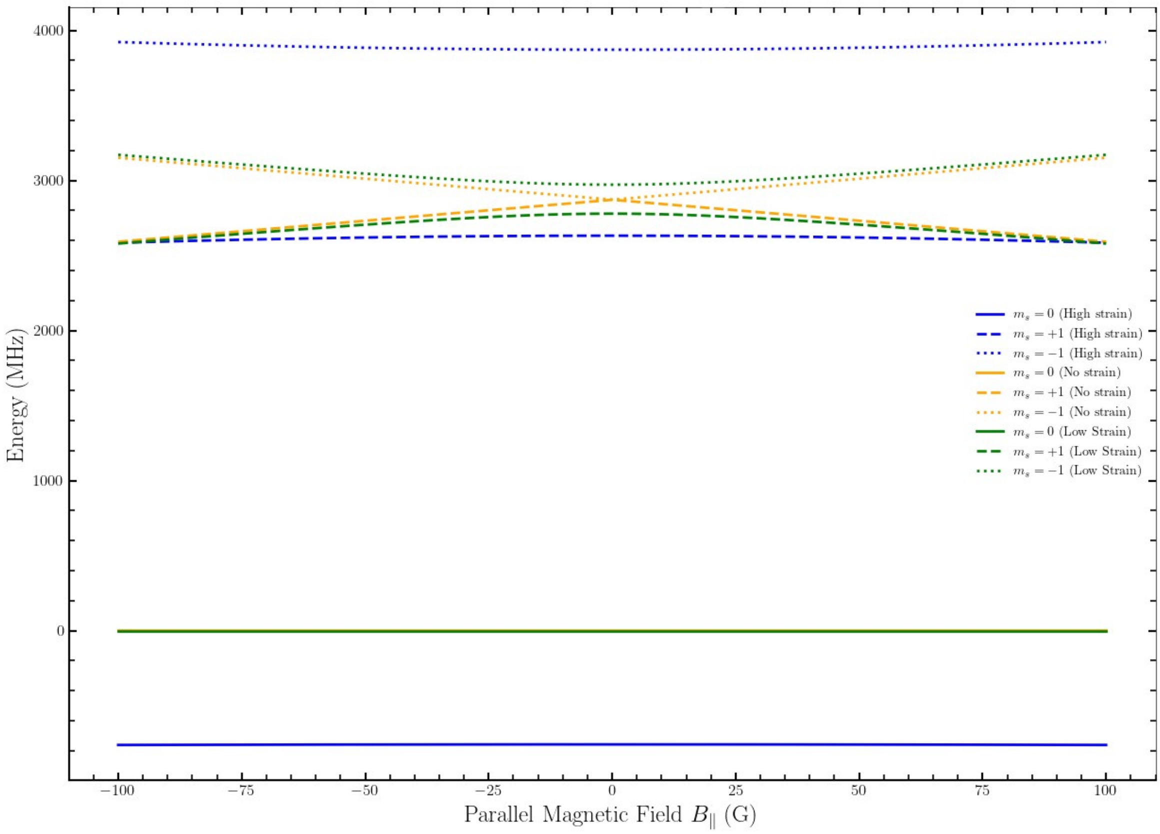

Figure 3: *Simulated energy levels of the NV center spin sublevels ($m_s = 0, \pm 1$) under varying parallel magnetic field $B_{\|}$ for three strain regimes: high strain (blue), low strain (green), and no strain (orange). The Zeeman splitting causes the $m_s = \pm 1$ levels to diverge linearly with $B_{\|}$, while strain-induced splitting is evident even at $B_{\|} = 0$.*

At zero parallel strain, the spin sublevels $m_s = \pm 1$ remain degenerate at $B_{\|} = 0$ and diverge symmetrically as the field increases due to Zeeman splitting. This is expected as the Hamiltonian is predominantly the zero-field splitting term D_0 and a linear Zeeman term $\gamma_{\mathrm{NV}} B_{\|}$ (Doherty et al., 2013). When a parallel strain is introduced (high strain $\sigma_x = 1GHz$, low strain $\sigma_x = 0.1GHz$.), the degeneracy at zero magnetic field is lifted, the $m_s = \pm 1$ levels exhibit an energy separation that increases with strain amplitude. This behaviour is expected as it arises from the strain-induced off-diagonal elements in the Hamiltonian, which couple the $m_s = \pm 1$ states (Maze et al., 2011; Barfuss et al., 2015).

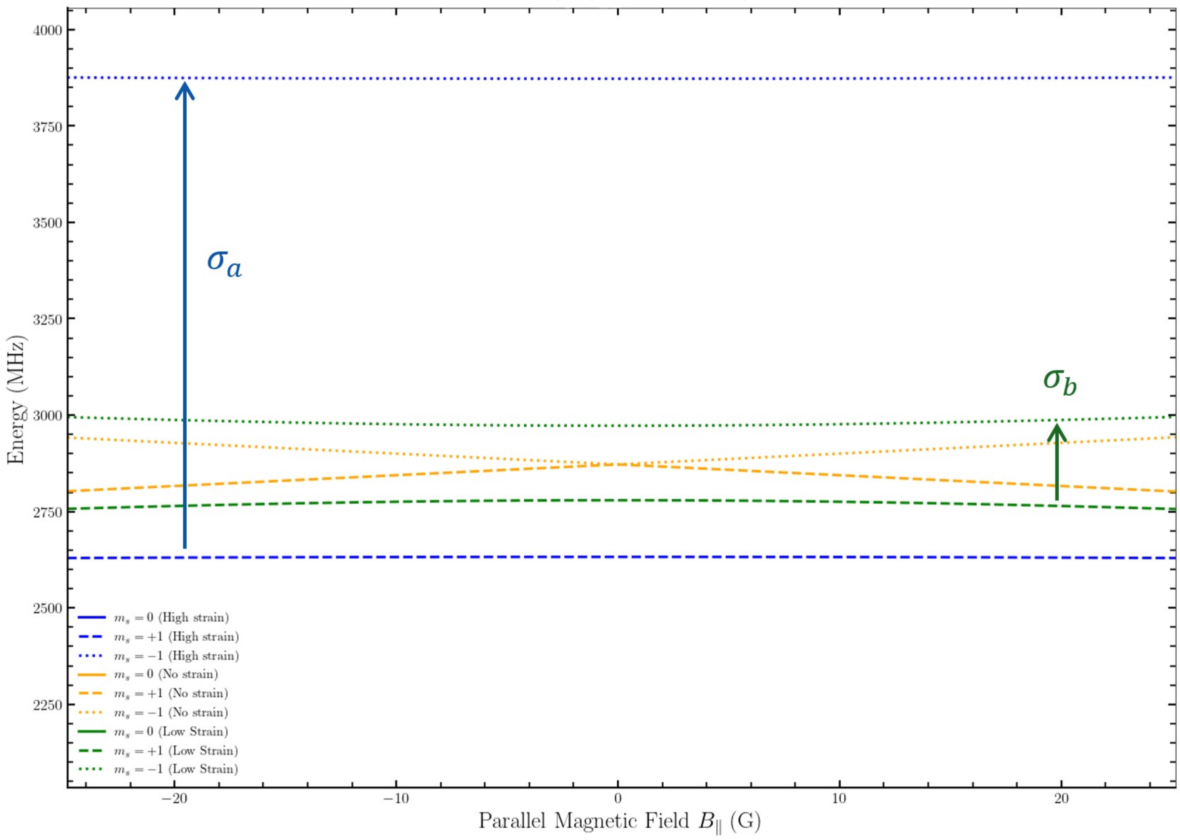

Figure 4: *Zoomed-in view of the same data as Figure 3, emphasizing strain-induced splitting between $m_s = \pm 1$ levels. The magnitude of the splitting is annotated as $\sigma_a = 1GHz$ for high strain and $\sigma_b = 0.1GHz$ for low strain, highlighting how increased strain amplitude correlates to the separation between energy level.*

Figure 4 quantifies this splitting as σ_a and σ_b for high and low strain conditions, the difference in energy levels corresponds to the magnitude of each strain. The energy gap observed at $B_{\|} = 0$ serves as a critical indicator of the strain-coupling strength and the presence of anti-crossings. The low strain term generates the conditions where energy levels approach each other closely but avoid crossing, therefore exhibiting anti-crossing behaviour. This anti-crossing has a range of magnetic field values from $-10G$ - $10G$, this anti-crossing range is used to determine the resonance conditions for the strain-mediated spin transitions. These findings demonstrate that coherent control of the NV centers spin states may be achieved using strain alone, in the absence of external radiofrequency fields.

Furthermore, the results suggest that strain-tunable energy splitting could be harnessed for remote sensing applications where traditional microwave/radio-wave delivery is impractical. Future research could focus on the implementation of mechanical or photothermal transducers to generate high-frequency strain fields, and the potential integration of these effects with ODMR protocols for quantum sensing (Barson et al., 2017; Teissier et al., 2014b).

The time evolution of the NV centers spin populations revealed distinct strain driven transitions between each spin manifold. The magnitude of the transitions were contingent on both the amplitude and frequency of the applied strain. The system was initialized in the $m_s = 0$ eigenstate, corresponding to full occupation of eigenstate 0 at $t = 0$. We applied a constant parallel magnetic field close to the anti-crossing ($10G$) observed in Figure 4. We discovered that a minimum strain amplitude of $2e8Hz$ and strain frequency of $2e8Hz$ was required at $10G$ to observe any perturbation effects. Using these minimum values the evolution of the eigenstate populations was observed over time and the strain frequency was varied until resonant conditions were observed.

In Figure 5a, the absence of applied strain ($\sigma_{\text{amp}} = 0$, $\sigma_{\text{freq}} = 0$), the system exhibits strictly time-independent behaviour. The population remains fully localized in eigenstate 0 throughout the simulation interval, with eigenstates 1 and 2 ($m_s = \pm 1$) remaining unpopulated. This serves as a baseline condition, verifying that in the absence of time-dependent perturbations, the NV center Hamiltonian preserves its initial eigenstate populations, as expected from unitary evolution under a static Hamiltonian.

In Figure 5b, low-frequency driving ($\sigma_{\text{amp}} = 2 \times 10^8$ Hz, $\sigma_{\text{freq}} = 2 \times 10^8$ Hz) demonstrates weak perturbative effects. Small-amplitude oscillations emerge in the population of eigenstate 0, but the overall occupation remains relatively undisturbed. This indicates that the strain frequency lies off-resonance with respect to the eigenenergy separations, however, does display that strain can induce some effect on eigenstate populations. Although the strain-induced terms introduce a time-dependent perturbation to the Hamiltonian, the driving frequency is insufficient to induce coherent transitions between the spin sublevels.

In Figure 5c, a higher driving frequency ($\sigma_{\text{amp}} = 2 \times 10^8$ Hz, $\sigma_{\text{freq}} = 5 \times 10^8$ Hz). In this case, substantial and periodic population transfer is observed between all three eigenstates, we observed Rabi-like oscillations between the different spin states. The population in eigenstate 0 exhibits significant oscillations, dipping to approximately 0.5 before partial recovery. Eigenstates 1 and 2 ($m_s = \pm 1$) show corresponding growth, with transient populations approaching 0.5, indicating coupling between the spin manifolds. The oscillatory behaviour indicates that the

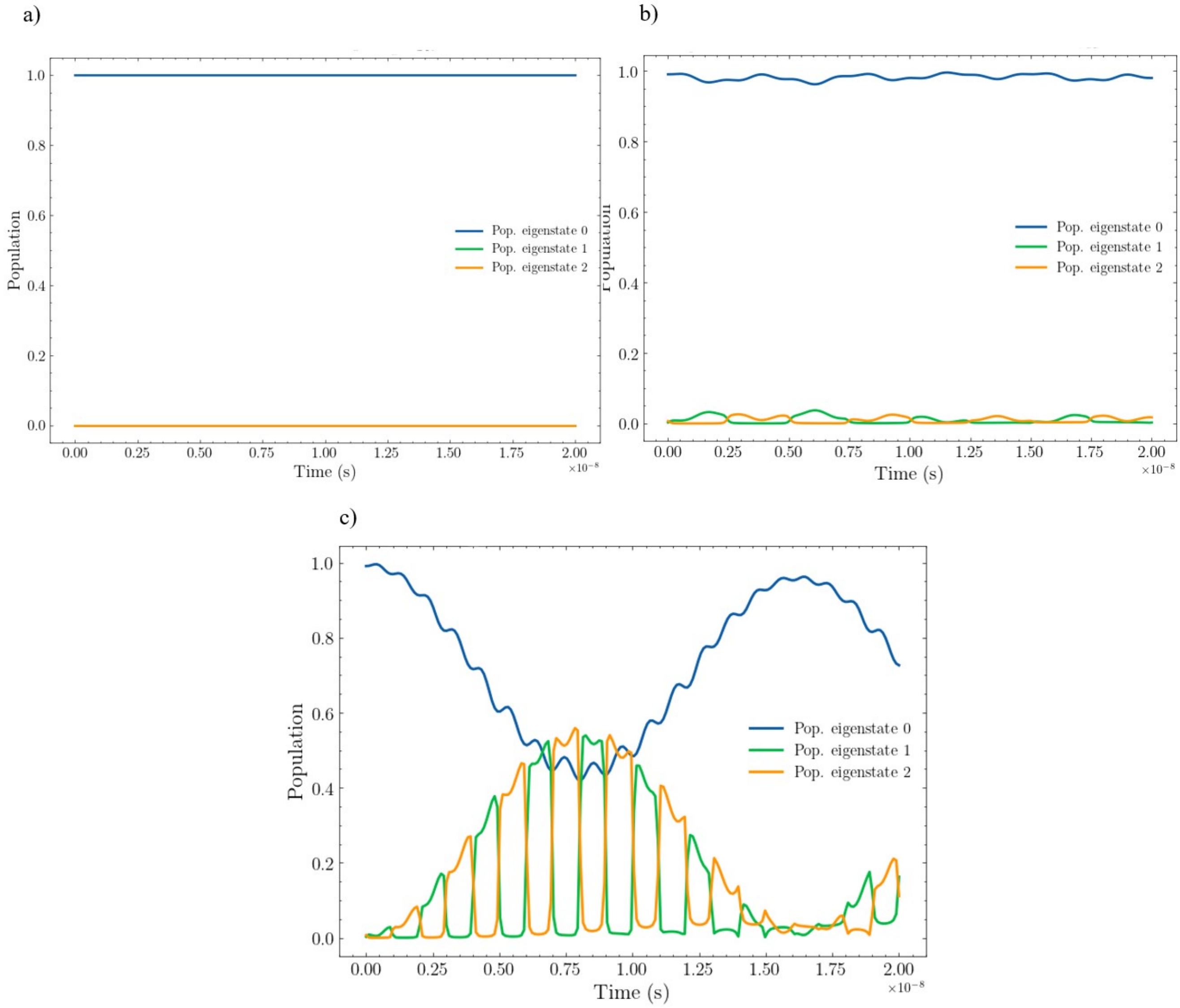

Figure 5: *Time evolution of NV center spin state populations under three different strain modulation conditions, all initialized in the $m_s = 0$ state. Panel (a) shows the unperturbed case with zero strain amplitude and frequency, where the population remains entirely in eigenstate 0. Panel (b) presents the dynamics with a strain amplitude of $2e8Hz$ and strain frequency of $2e8Hz$, resulting in small-amplitude oscillations and minimal population transfer. Panel (c) displays the effect of increasing the strain frequency to $5e8Hz$ (with the same amplitude), revealing pronounced coherent oscillations and significant population transfer among all three eigenstates, indicative of resonant strain driven coupling.*

strain frequency of ($\sigma_x = 5e8Hz$) approaches resonance, where the modulation frequency closely matches the energy spacing between the spin sublevels. This resonant coupling enables efficient and varying population transfer between spin states, demonstrating the feasibility of strain driven spin manipulation without requiring a MW or RF field (stress terms).

These dynamics are governed by the time-dependent off-diagonal components in the Hamiltonian, introduced via the $\sigma_x(t)$ and $\sigma_y(t)$ strain terms. These terms mediate coupling between the spin-1 states, allowing for transitions that are otherwise forbidden in the absence of transverse perturbations. The transition probability is maximized under resonant conditions, where the strain driving frequency matches the energy splitting induced by the zero-field and Zeeman terms. The high-frequency regime, in particular, demonstrates the viability of phonon-driven spin transitions through dynamic modulation of the crystal field environment.

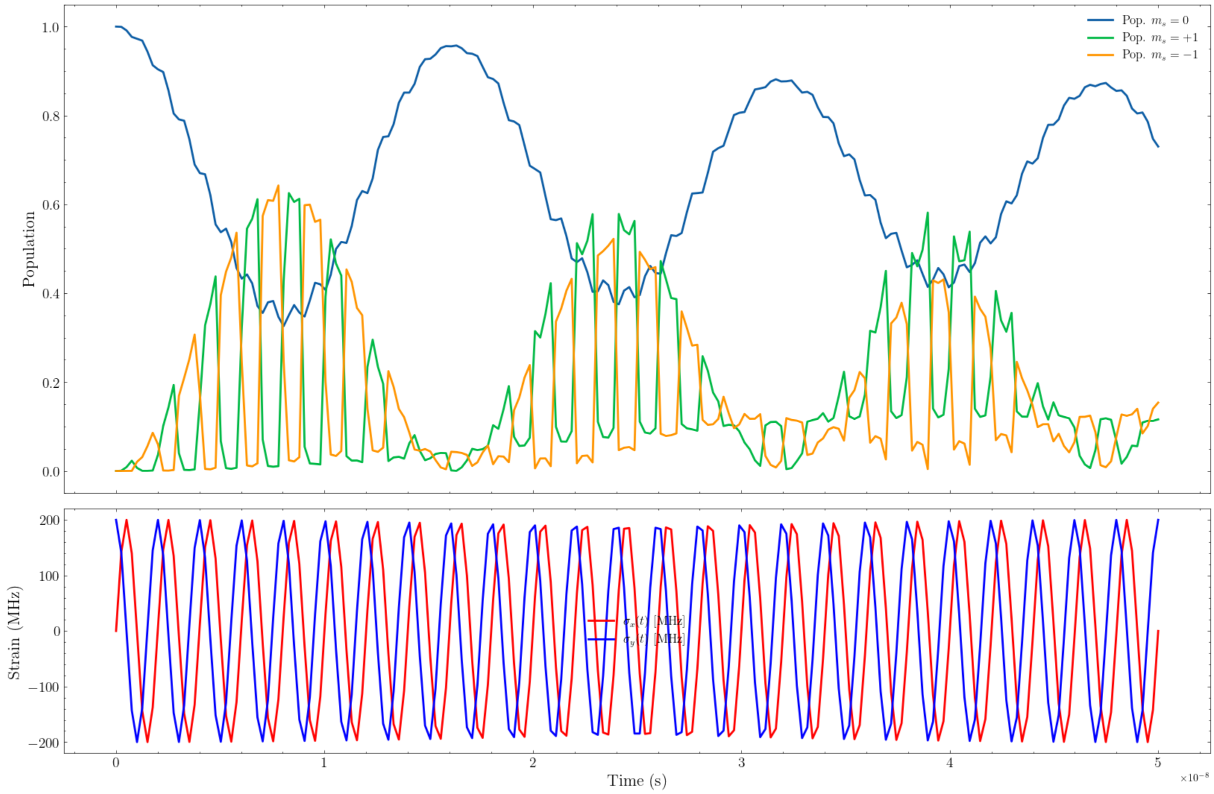

Figure 6: *Population dynamics of the NV center spin states under resonant strain modulation (strain amplitude = $2e8Hz$, strain frequency = $5e8Hz$) over an extended time scale. The upper panel shows the time evolution of the populations in the $m_s = 0$, $m_s = +1$ and $m_s = -1$ states, initialized with all population in $m_s = 0$. The lower panel displays the corresponding time-dependent strain components $\sigma_x(t)$ and $\sigma_y(t)$. The large-amplitude, coherent oscillations in the populations confirm resonant coupling between the spin states under these strain conditions.*

Figure 6 displays the extended time analysis of the population dynamics at the near-resonant strain frequency ($\sigma_{amp} = 2e8Hz$,$\sigma_{freq} = 5e8Hz$) observed in Figure 5c. By extending the duration from $2e-8$ to $5e-8$ seconds we confirmed the persistent and coherent nature of the strain-induced population dynamics over extended time scales. The oscillations in eigenstate 0 exhibit strong stability, maintaining large amplitude variations between 0.5 and 0.95 throughout the simulation period. The population dynamics show no evidence of decoherence and only minor variations in amplitude over the extended observation window, indicating that the strain-induced transitions maintain phase coherence. The eigenstate 1 and eigenstate 2 ($m_s = \pm 1$), populations exhibit oscillations with comparable amplitudes to eigenstate 0 ($m_s = 0$), reaching peak values of approximately 0.5-0.6, confirming efficient population transfer between all three spin states. Notably, the oscillation pattern reveals a complex multi-frequency structure, with both rapid oscillations superimposed on slower envelope modulations. This behaviour suggests the presence of multiple coupling pathways and beating effects between different transition channels, consistent with the three-level nature of the spin-1 system under strain modulation.

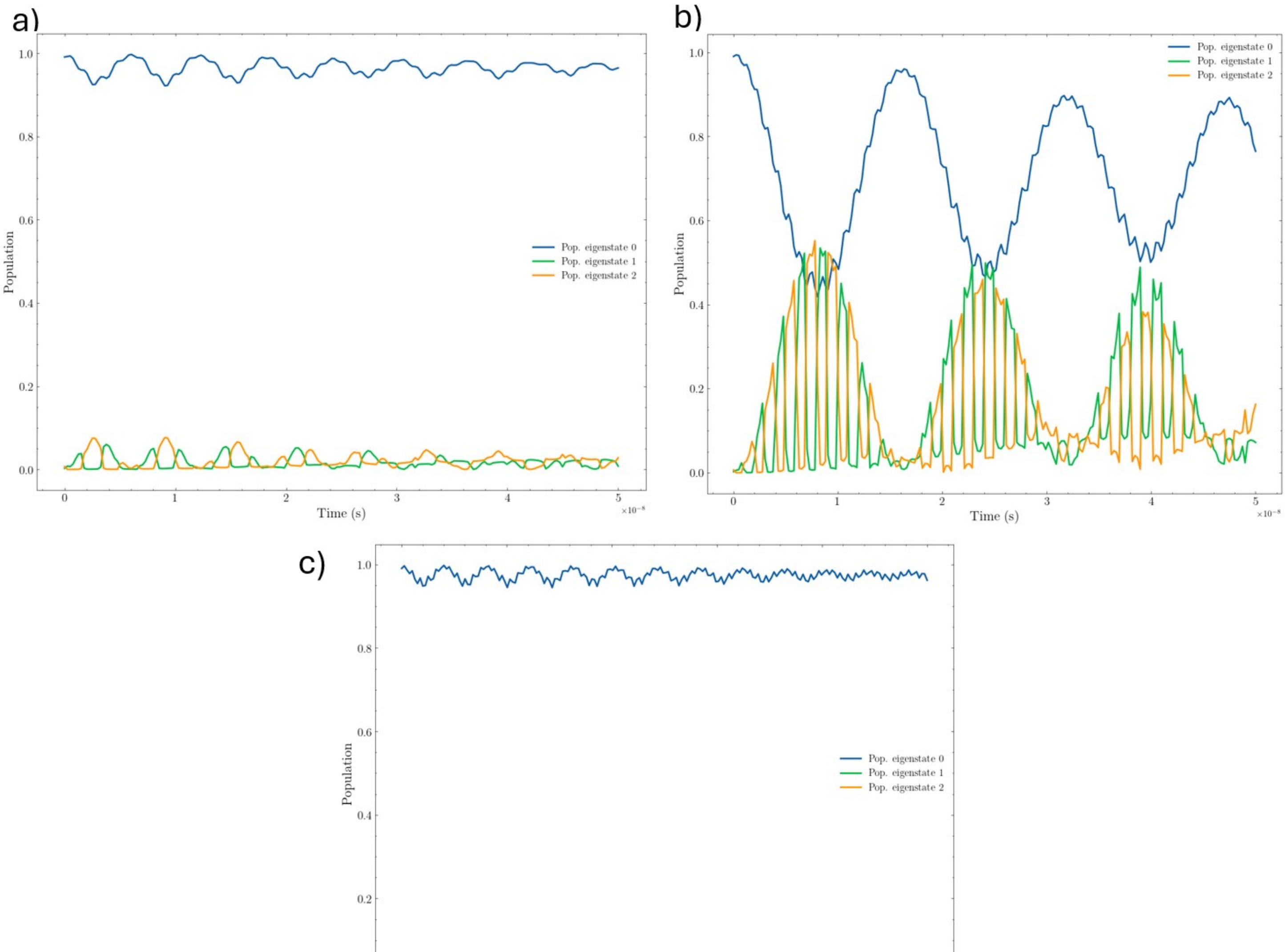

Figure 7: *Comparison of NV center population dynamics for strain modulation frequencies at, below, and above resonance. Panel (a) shows the population evolution at sub-resonant frequency and (c) at above-resonant frequency (7e8Hz), all with strain amplitude fixed at 2e8Hz. Significant population transfer and large oscillation amplitudes are observed only at the resonant frequency, demonstrating the frequency selectivity and efficiency of strain driven spin transitions*

Figure 7 shows a frequency comparison study provides to provide definitive evidence for the resonant nature of the strain-spin coupling mechanism. Three distinct frequency regimes are compared: the resonant frequency ($5e8Hz$), a sub-resonant frequency ($3e8Hz$), and an above-resonant frequency ($7e8Hz$), each differing by $\pm 2e8Hz$ from the optimal value.

At the resonant frequency Figure 7b, the system exhibits the dramatic population oscillations observed previously, with eigenstate 0 population dropping to approximately 0.5 while eigenstates 1 and 2 show substantial and sustained population transfer. The oscillation amplitudes remain large throughout the simulation, confirming efficient energy exchange between the strain field and the spin system.

In contrast, both the sub-resonant ($3e8Hz$, left panel) and super-resonant ($7e8Hz$ Hz, right panel) cases show dramatically reduced population transfer. The eigenstate 0 population remains close to unity with only small-amplitude oscillations visible, while eigenstates 1 and 2 exhibit minimal population accumulation. This frequency dependence demonstrates a sharp resonance condition, with detuning by $\pm 2e8Hz$ sufficient to suppress the strain-induced transitions almost entirely.

The observed frequency selectivity provides crucial insight into the underlying strain-spin coupling mechanism. The resonant frequency of 500 MHz falls within the range expected for transitions between the spin sublevels under the applied conditions, suggesting that the strain field effectively drives coherent Rabi oscillations between the spin states. The persistence of coherent oscillations over extended time scales indicates that the strain coupling mechanism preserves the quantum coherence of the spin system. This behaviour is essential for practical applications, as it demonstrates that strain-induced control can maintain the quantum properties that make NV centers valuable for sensing and quantum information applications. The sharp frequency dependence observed in the comparison study has important implications for experimental implementation. The narrow resonance window suggests that precise frequency control will be required to achieve efficient strain driven spin manipulation but also indicates that the technique offers excellent selectivity for addressing specific transitions in multi-level systems.

The observed Rabi-like oscillations confirm that strain-induced transitions can replicate key features of conventional ODMR behaviour. Future investigations should aim to quantify transition rates across a broader frequency spectrum, explore multi-photon coupling effects at higher strain amplitudes, and assess decoherence and damping mechanisms under realistic experimental noise conditions. These results collectively establish that time-varying strain represents a viable alternative to magnetic resonance for NV center spin control, with the potential for achieving coherent manipulation while maintaining the advantageous properties of the quantum system.

In this section, we outline the primary sources of uncertainty in the computational model and reference the specific code modules responsible for parameter definition, numerical integration, and output visualisation. Unfortunately, quantification of the uncertainty was unable to be completed at this stage.

Numerical integration error arises from the use of the solveivp function in SciPy, specifically with the Runge-Kutta method of order 5(4) (RK45), as implemented in the timedependentpopulation method of the NVHamiltonian class. The integration tolerances are set to rtol=$1e-8$ and atol=$1e-8$, which are stringent but still introduce a finite numerical error. This error could be estimated by comparing results at different step sizes or by using higher-order solvers for benchmarking in the future. The simulation outcomes depend sensitively on physical parameters such as the zero-field splitting D_0, strain amplitude, strain frequency, and magnetic field components, all defined at the top of the code and passed as arguments to the Hamiltonian builder (buildhamiltonian). Small uncertainties or systematic errors in these parameters, arising from experimental measurement or literature values, propagate through the simulation and affect resonance conditions and population transfer rates.

The Hamiltonian implemented in the code (buildhamiltonian) includes only the dominant terms: zero-field splitting, Zeeman effect, Hyperfine splitting (as a constant) and linear strain coupling. Higher-order effects, nuclear spin effects, decoherence, and environmental noise are neglected, which may lead to discrepancies when comparing with experimental data. All simulations are initialized with the population entirely in the $m_s = 0$ state (see the initialprobs argument in timedependentpopulation, (see Appendix B)). In a real experiment, imperfect initialization or thermal mixing would introduce uncertainty in the initial population distribution, potentially leading to deviations in predicted dynamics. The simulation time grid is defined by nsteps and tmax (see all simulation functions, e.g., timedependentpopulation, timevaryingstrainsimulation). Coarse discretization can miss fast oscillations or introduce aliasing, while excessively fine grids increase computational cost and may accumulate floating-point errors.

To assess sensitivity, parameter sweeps could be performed by systematically varying strain amplitude, frequency, or magnetic field and observing the resulting changes in population dynamics. This is supported by the modular structure of the simulation functions (see timedependentpopulation and timevaryingstrainsimulation). For each output observable (e.g., population in a given eigenstate), uncertainties can be propagated using Monte Carlo simulations: repeatedly sampling input parameters from their estimated distributions and analyzing the spread in the resulting population traces. The code's modular structure allows for straightforward extension to include decoherence, temperature effects, or experimental noise, which would provide a more comprehensive uncertainty for future work.

Conclusion

This study demonstrates that coherent spin manipulation in NV centers can be achieved through time-varying strain alone, without reliance on conventional RF or microwave excitation. By simulating the spin dynamics under photothermal strain, we identified a clear resonance condition at which the system undergoes efficient population transfer between spin states. These Rabi-oscillations indicate that phonon-mediated transitions replicate the fundamental features of ODMR. The sharp frequency selectivity and long-lived coherence observed highlight the potential of this method for quantum sensing, particularly in RF-inaccessible environments such as biological tissue or conductive media. While this computational model simplifies certain physical aspects, including decoherence, higher-order hyperfine interactions, and environmental noise, the results establish a viable pathway for RF-free quantum control. Future work should extend these simulations to include decoherence mechanisms, explore multi-photon transitions, and experimentally validate the resonance conditions predicted here.

Scientific Mastery

This project has been a challenging and exciting experience that expanded my understanding of quantum systems, computational physics, and scientific practice. When the topic of phonon-driven spin manipulation in NV centers was first introduced to me by Professor Andrew Greentree, I was presented with an open-ended research question: Can photothermal excitation be used as a viable method of quantum spin control in diamond NV centers? At the time, I had no direction, only the fragments of prior coursework, academic papers, and the unfamiliar concepts of quantum magnetometry systems.

To approach this challenge, I began by immersing myself in the literature on NV centers, optically detected magnetic resonance (ODMR), and strain-induced spin transitions. I attended weekly meetings with Professor Greentrees research group, where I was introduced to the work of research fellows and PhD candidates. These sessions proved invaluable, not only for the technical knowledge I absorbed but also for the motivation they provided. Being surrounded by researchers actively solving real world problems inspired a sense of responsibility and determination for me to contribute meaningfully to my project.

The first major milestone was the derivation of the NV center spin Hamiltonian under dynamic strain. This derivation was performed by hand over several days. Sitting in the library, I had nothing but a pen, whiteboard and my reference materials on spin-1 systems under strain and magnetic field interactions. The final Hamiltonian incorporated zero-field splitting, Zeeman and hyperfine interactions, and strain-induced off-diagonal terms. This task required synthesizing theoretical physics concepts with applied mathematical formalism, something I had not done at this level of complexity before. I documented this derivation in LaTeX, which I learned from scratch to produce professional-grade documents suitable for academic publishing. This skill became fundamental not only for the project report but also for my work in other physics courses, where I now consistently use LaTeX to present my results in reports.

Following the theoretical work, I transitioned into developing a simulation in Python that could solve the time-dependent Schrdinger equation (TDSE) for the NV center Hamiltonian. This phase proved to be the most technically demanding. I revised and rebuilt the simulation over seven major iterations, integrating tools like NumPy, SciPy, and Matplotlib, while balancing mathematical rigor with computational stability. By the end of the project I had programmed and developed seven iterations of my projects code. I taught myself advanced Python programming techniques, adaptive ODE solvers, and eigenvalue tracking methods to simulate dynamic population transfer between spin states under oscillating strain fields.

Throughout the development process, I encountered numerous setbacks including numerical instability, poor convergence at high frequencies, and errors arising from Hamiltonian dimensionality when extending to hyperfine interactions. Overcoming these issues required persistence, guidance from my supervisors, and many hours of independent problem-solving. By the end, I had produced a working simulation that could model energy level evolution, identify anti-crossing regimes, and track population dynamics under resonant strain conditions.

The skills demonstrated in this project are broad and interconnected. I learnt analytical derivation, numerical simulation, data visualization, technical communication, and self-directed learning. More importantly, this work has deepened my conceptual understanding of spin-based quantum systems and solidified my confidence in approaching complex research problems independently. I now feel equipped not only with the tools of quantum theory and computation but with the mindset required to contribute to the broader field of quantum science and engineering.

References

Victor M. Acosta. Direct spin phonon coupling with nitrogen vacancy centers in diamond. https://spie.org/news/5348directspinphononcouplingwithnitrogenvacancycentersindiamond, 2014. URL https://spie.org/news/5348directspinphononcouplingwithnitrogenvacancycentersindiamond. SPIE Newsroom.

A. Barfuss, J. Teissier, E. Neu, A. Nunnenkamp, and P. Maletinsky. Strong mechanical driving of a single electron spin. *Nature Physics*, 11(10):820–824, 2015. doi: 10.1038/nphys3411. URL https://doi.org/10.1038/nphys3411.

Arne Barfuss, Daniil T. Smirnov, and Patrick Maletinsky. Strain engineering and control of quantum states in solid state spin systems. *Applied Physics Reviews*, 11(1):011301, 2024. doi: 10.1063/5.0180354.

Maxwell S. J. Barson, Pallavi Peddibhotla, Preeti Ovartchaiyapong, Kumaravelu Ganesan, Lindsay A. Taylor, Max Gebert, Zachary Mielens, Bryan A. Myers, Ania C. Bleszynski Jayich, and David J. Reilly. Nanomechanical sensing using spins in diamond. *Nano Letters*, 17(3): 1496–1503, 2017. doi: 10.1021/acs.nanolett.6b05102. URL https://doi.org/10.1021/acs.nanolett.6b05102.

Marcus W Doherty, Neil B Manson, Paul Delaney, Fedor Jelezko, Jørg Wrachtrup, and Lloyd C L Hollenberg. The nitrogen vacancy colour centre in diamond. *Physics Reports*, 528(1):1 to 45, 2013.

D. A. Golter, T. Oo, M. Amezcua, K. A. Stewart, and H. Wang. Coupling a surface acoustic wave to an electron spin in diamond via a dark state. *Physical Review X*, 6(4):041060, 2016. doi: 10.1103/PhysRevX.6.041060.

Fedor Jelezko and Jørg Wrachtrup. Single defect centres in diamond: A review. *Physica Status Solidi (a)*, 203(13):3207to3225, 2006.

Erik R. MacQuarrie, Thomas A. Gosavi, Nathalie R. Jungwirth, Samuel A. Bhave, and Gregory D. Fuchs. Mechanical spin control of nitrogen vacancy centers in diamond. *Physical Review Letters*, 111(22):227602, 2013. doi: 10.1103/PhysRevLett.111.227602.

Evan R. MacQuarrie, Tanay A. Gosavi, Alec M. Moehle, Nicholas R. Jungwirth, Sunil A. Bhave, and Gregory D. Fuchs. Mechanical spin control of nitrogen vacancy centers in diamond. *Phys. Rev. B*, 92(22):224419, 2015.

S. Maity, G. Calusine, H. Kim, L. R. Sletten, R. Riedinger, R. J. Schoelkopf, A. Melville, K. Rostem, M. Hatridge, M. H. Devoret, A. N. Cleland, and D. I. Schuster. Coherent acoustic control of a single silicon vacancy spin in diamond. *PRX Quantum*, 5(3):030336, 2024. doi: 10.1103/PRXQuantum.5.030336. URL https://doi.org/10.1103/PRXQuantum.5.030336.

J. R. Maze, A. Gali, E. Togan, Y. Chu, A. Trifonov, E. Kaxiras, and M. D. Lukin. Properties of nitrogen-vacancy centers in diamond: the group theoretic approach. *New Journal of Physics*, 13(2):025025, 2011. doi: 10.1088/1367-2630/13/2/025025. URL https://doi.org/10.1088/1367-2630/13/2/025025.

P. Ovartchaiyapong, K. W. Lee, B. A. Myers, and A. C. Bleszynski Jayich. Dynamic strain mediated coupling of a single diamond spin to a mechanical resonator. *Physical Review Letters*, 111(22):227602, 2013. doi: 10.1103/PhysRevLett.111.227602. URL https://doi.org/10.1103/PhysRevLett.111.227602.

Benjamin Smeltzer, Jordan McIntyre, and Lilian Childress. Robust control of individual nuclear spins in diamond. *Phys. Rev. A*, 80(5):050302, 2009.

J. Teissier, A. Barfuss, P. Appel, E. Neu, and P. Maletinsky. Strain coupling of a nitrogen vacancy center spin to a diamond mechanical oscillator. *Phys. Rev. Lett.*, 113(2):020503, 2014a.

J. Teissier, A. Barfuss, P. Appel, E. Neu, and P. Maletinsky. Strain coupling of a nitrogen-vacancy center spin to a diamond mechanical oscillator. *Physical Review Letters*, 113(2): 020503, 2014b. doi: 10.1103/PhysRevLett.113.020503. URL https://doi.org/10.1103/PhysRevLett.113.020503.

Appendix

A Derivation of the Hamiltonian

Spin 1 operators: ($\hbar = 1$)

$$S_x = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \tag{5}$$

$$S_x^2 = \frac{1}{2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \tag{6}$$

$$S_y = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{bmatrix}, \tag{7}$$

$$S_y^2 = \frac{1}{2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \tag{8}$$

$$S_z = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \tag{9}$$

$$S_z^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{10}$$

Hamiltonian:

$$\begin{aligned} H_{NV} = &\left(D_0 + \epsilon_{\parallel}\sigma_{\parallel}\right) S_z^2 + \gamma_{NV} B_{\parallel} S_z + \gamma_{NV} B_{\perp} S_x ... \\ &+ \epsilon_{\perp}\sigma_x (S_x^2 - S_y^2) + \epsilon_{\perp}\sigma_y (S_x S_y + S_y S_x) \end{aligned} \tag{11}$$

List of variables/terms:

$\epsilon_{\parallel} \rightarrow$ N/A

Variable	Description	Magnitude
D_0	zero-field splitting term	2.87 GHz
γ_{NV}	NV centre gyromagnetic ratio	2.8 MHz/G
$\epsilon_\perp$	stress coupling constants	0.03 MHz/MPa

$B_\parallel \to$ N/A

$B_\perp \to$ N/A

$\sigma_\parallel$ σ_x σ_y (stress terms)

$$(\gamma_{NV} B_\parallel S_z + \gamma_{NV} B_\perp S_x) \to \text{Zeeman splitting term}$$

Matrix Working

Operator Form

$$\begin{aligned} H_{\mathrm{NV}} &= \left(D_0 + \epsilon_\parallel \sigma_\parallel\right) S_z^2 + \gamma_{\mathrm{NV}} B_\parallel S_z + \gamma_{\mathrm{NV}} B_\perp S_x \\ &\quad + \epsilon_\perp \sigma_x (S_x^2 - S_y^2) + \epsilon_\perp \sigma_y (S_x S_y + S_y S_x) \end{aligned} \tag{12}$$

Matrix Representation (Spin-1 basis)

$$\begin{aligned} H_{\mathrm{NV}} = \sum_i \mathcal{O}_i = &\underbrace{(D_0 + \epsilon_\parallel \sigma_\parallel) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Zero-field splitting}} + \underbrace{\gamma_{\mathrm{NV}} B_\parallel \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}}_{\text{Axial Zeeman}} \\ &+ \underbrace{\frac{\gamma_{\mathrm{NV}} B_\perp}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{\text{Transverse Zeeman}} + \underbrace{\epsilon_\perp \sigma_x \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}}_{\text{Strain } \sigma_x} + \underbrace{i\epsilon_\perp \sigma_y \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}}_{\text{Strain } \sigma_y} \end{aligned}$$

Explicit 33 Matrix

$$H_{\mathrm{NV}} = \begin{bmatrix} D_0 + \epsilon_\parallel \sigma_\parallel + \gamma_{\mathrm{NV}} B_\parallel & \frac{\gamma_{\mathrm{NV}} B_\perp}{\sqrt{2}} - i\epsilon_\perp \sigma_y & \epsilon_\perp \sigma_x \\ \frac{\gamma_{\mathrm{NV}} B_\perp}{\sqrt{2}} + i\epsilon_\perp \sigma_y & 0 & \frac{\gamma_{\mathrm{NV}} B_\perp}{\sqrt{2}} - i\epsilon_\perp \sigma_y \\ \epsilon_\perp \sigma_x & \frac{\gamma_{\mathrm{NV}} B_\perp}{\sqrt{2}} + i\epsilon_\perp \sigma_y & D_0 + \epsilon_\parallel \sigma_\parallel - \gamma_{\mathrm{NV}} B_\parallel \end{bmatrix} \tag{13}$$

Eigenvalue Equation

$$\det(H_{\text{NV}} - E \parallel I) = 0 \tag{14}$$

$$= \det \begin{pmatrix} D_0 + \epsilon_{\parallel}\sigma_{\parallel} + \frac{\gamma_{NV}B_{\parallel}}{\sqrt{2}} - \epsilon & \frac{\gamma_{NV}B_{\parallel}}{\sqrt{2}} - i(\epsilon_{12}\sigma_{12}) \\ \frac{\gamma_{NV}B_{\parallel}}{\sqrt{2}} + i(\epsilon_{12}\sigma_{12}) & D_0 + \epsilon_{\parallel}\sigma_{\parallel} - \frac{\gamma_{NV}B_{\parallel}}{\sqrt{2}} - \epsilon \end{pmatrix} = 0$$

Let:

$$\begin{aligned} a &= D_0 + \epsilon_{\parallel}\sigma_{\parallel} \\ b &= (\frac{\gamma_{NV}B_{\perp}}{\sqrt{2}} - i\epsilon_{\perp}\sigma_y) \\ b^* &= (\frac{\gamma_{NV}B_{\perp}}{\sqrt{2}} + i\epsilon_{\perp}\sigma_y) \\ c &= \epsilon_{\perp}\sigma_x \\ d &= D_0 + \epsilon_{\parallel}\sigma_{\parallel} - \gamma_{NV}B_{\parallel} \end{aligned}$$

$$det(H_{NV} - EI) = \begin{pmatrix} a-E & b & c \\ b* & -E & b \\ c & b^* & d-E \end{pmatrix}$$

$$\det = (a-E)[(-E)(d-E) - bb^*] + b[(b^*)(d-E) - bc] + c[(b^*)(b^*) - (-E)(c)]$$

$$\det = (a-E)(-Ed + E^2 - (b^*)^2) - bb^*d + bb^*E + b^2c + c(b^*)^2 + c^2E$$

Considering the system without strain, we set the strain to zero: $\epsilon_{\perp} = 0$

$$\begin{aligned} b &= (\frac{\gamma_{NV}B_{\perp}}{\sqrt{2}}) \\ b^* &= b \\ c &= 0 \\ \sigma_y &= 0 \end{aligned}$$

$$det(H_{NV} - EI) = \begin{pmatrix} a-E & b & 0 \\ b & -E & b \\ 0 & b & d-E \end{pmatrix}$$

New determinant of new matrix:

$$\begin{aligned} det = (a-E)[(-E)(d-E) - b^2] - b[b(d-E)] + 0 \\ = (a-E)[E^2 - Ed - b^2] - b[bd - bE] \\ = aE^2 - aEd - ab^2 - E^3 + E^2d + Eb^2 - b^2d - b^2E \\ = E^3 + (a+d)E^2 + (b^2 + b^2 + ad)E - ab^2 - b^2d \end{aligned}$$

Now considering only the diagonal terms:

$$a \approx D_0 + \gamma_{NV} B_{\|}$$
$$d \approx D_0 - \gamma_{NV} B_{\|}$$

Substitute into $(a-E)(d-E)(-E)$:

$$(D_0 - \gamma_{NV} B_{\|} - E)(D_0 - \gamma_{NV} B_{\|})(-E)$$

Let $x = E$:

$$\begin{aligned} det(H_{NV} - EI) = -(D_0 + \gamma_{NV} B_{\|} - x)(D_0 - \gamma_{NV} B_{\|} - x)(x) \\ det = -x[(D_0 - x)^2 - (\gamma_{NV} B_{\|})^2] \\ = -x[D_0^2 - x^2 - 2D_0 x - (\gamma_{NV} B_{\|})^2] \\ = -x^3 + 2D_0 x^2 - D_0^2 x - \gamma_{NV}^2 B_{\|}^2 x \\ = -x^3 + 2D_0 x^2 - x(D_0 - \gamma_{NV} B_{\|})^2 \end{aligned}$$

Substitute $x = E$

$$-E^3 + 2D_0 E^2 - E(D_0 - \gamma_{NV} B_{\|})^2 = 0$$

Therefore, the unperturbed spectrum, where $\epsilon_\perp = 0$ is:

$$-E^3 + 2D_0E^2 - E(D_0 - \gamma_{NV}B_\parallel)^2 = 0$$
$$E(E^2 - 2D_0E + D_0^2 - \gamma_{NV}^2B_\perp^2) = 0$$

Using the quadratic equation:

$$E = \frac{2D_0 \pm \sqrt{(2D_0)^2 - 4(D_0^2 - \gamma_{NV}^2B_\parallel^2)}}{2}$$
$$E = D_0 \pm \gamma_{NV}B_\parallel$$

The Eigenvalues are:

$$E_1 = 0$$
$$E_2 = D_0 + \gamma_{NV}B_\parallel$$
$$E_3 = D_0 - \gamma_{NV}B_\parallel$$

Substituting the values for the zero-field splitting term and gyromagnetic ratio:

$$E_1 = 0$$
$$E_2 = (2.87)GHz + (2.8)MHz/G * B_\parallel$$
$$E_3 = (2.87)GHz - (2.8)MHz/G * B_\parallel$$

Without perturbations, the eigenvalues E_2 and E_3 are functions of $B_\parallel$.

B NV Simulator Code

Python Code:

Listing 1: NVSimulator Class Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import eigh  # For Hermitian matrix diagonalization
```

```
from matplotlib.gridspec import GridSpec  # For arranging subplots
from scipy.integrate import solve_ivp  # To solve ODEs (Schr dinger
    equation)
from scipy.signal import argrelextrema  # To find local extrema
# Optional science plot style
# import scienceplots
# plt.style.use('science')

# Constants (in Hz)
D0 = 2.87e9        # Zero-field splitting of NV center ground state (Hz)
gamma_NV = 2.8e6  # NV gyromagnetic ratio (Hz/G)

class NVHamiltonian:
    def __init__(self):
        self.verbose = False  # Enables detailed Hamiltonian printing
            for debugging

    def build_hamiltonian(self, B_parallel, B_perpendicular=0,
                          sigma_parallel=0, sigma_x=0, sigma_y=0,
                          A_iso=0, print_matrix=False):
        """
        Construct the NV center Hamiltonian with given parameters.
        All terms are in frequency units (Hz).
        """

        # Diagonal terms: affected by axial magnetic field, strain, and
            hyperfine interaction
        a = D0 + sigma_parallel + gamma_NV * B_parallel + A_iso
        d = D0 + sigma_parallel - gamma_NV * B_parallel + A_iso

        # Off-diagonal terms:
        #   b handles B_perpendicular + sigma_y (complex)
        #   c handles sigma_x (real)
        b = (gamma_NV * B_perpendicular / np.sqrt(2)) - 1j * sigma_y
        b_star = np.conj(b)
        c = sigma_x

        # Construct full 3x3 complex Hermitian Hamiltonian
        H = np.array([
```

```
            [a, b, c],
            [b_star, 0 + A_iso, b],
            [c, b_star, d]
        ], dtype=complex)

        # Optional: print the matrix if debug flag or parameter is set
        if print_matrix or self.verbose:
            self._print_hamiltonian(H, B_parallel, B_perpendicular,
                                    sigma_parallel, sigma_x, sigma_y,
                                       A_iso)

        return H

    def _print_hamiltonian(self, H, B_parallel, B_perpendicular,
                           sigma_parallel, sigma_x, sigma_y, A_iso):
        """
        Nicely formatted printout of the Hamiltonian matrix and its
           components.
        """
        print("\n" + "=" * 60)
        print("NV␣Center␣Hamiltonian␣Composition:")
        print(f"B_parallel␣=␣{B_parallel}␣G,␣B_perpendicular␣=␣{
           B_perpendicular}␣G")
        print(f"Strain:␣sigma_parallel␣=␣{sigma_parallel}␣Hz,␣sigma_x␣=␣
           {sigma_x}␣Hz,␣sigma_y␣=␣{sigma_y}␣Hz")
        print(f"Hyperfine:␣A_iso␣=␣{A_iso}␣Hz")
        print("\nFull␣Hamiltonian␣Matrix␣(Hz):")
        for row in H:
            print("[", end="")
            for val in row:
                if np.iscomplex(val):
                    print(f"{val.real:12.2f}{val.imag:+12.2f}j", end="␣"
                       )
                else:
                    print(f"{val.real:12.2f}{'␣'␣*␣12}", end="␣")
            print("]")
        print("=" * 60 + "\n")

    def diagonalize(self, H):
```

```
        """
        Diagonalize the Hamiltonian to get eigenvalues and eigenvectors.
        """
        evals, evecs = eigh(H)
        return evals, evecs

    def plot_energy_levels_vs_field(self, B_range=(-10, 10), n_points
       =1000,
                                    B_perpendicular=0, sigma_parallel=0,
                                    sigma_x=0, sigma_y=0, A_iso=0):
        """
        Plot NV center energy levels as a function of B_parallel field.
        Useful for visualizing Zeeman and strain-induced level shifts.
        """
        B_values = np.linspace(B_range[0], B_range[1], n_points)
        energies = []

        # Loop over magnetic field values and compute eigenvalues
        for B in B_values:
            H = self.build_hamiltonian(B, B_perpendicular,
               sigma_parallel,
                                      sigma_x, sigma_y, A_iso)
            evals, _ = self.diagonalize(H)
            energies.append(evals)

        energies = np.array(energies).T  # Shape: (3, n_points)
        spin_labels = ['$m_s=0$', '$m_s=+1$', '$m_s=-1$']

        # Plotting
        plt.figure(figsize=(12, 8))
        for i in range(3):
            plt.plot(B_values, energies[i]/1e6, label=spin_labels[i],
               linewidth=2)
        plt.title(f'NV␣Center␣Energy␣Levels␣vs␣$B_\\parallel$\n'
                  f'(Strain:␣$\\sigma_x={sigma_x:.1e}$␣Hz,␣$\\sigma_y={
                     sigma_y:.1e}$␣Hz,␣$B_\\perp={B_perpendicular:.1f}$␣
                     G)')
        plt.xlabel('Parallel␣Magnetic␣Field␣$B_\\parallel$␣(G)')
        plt.ylabel('Energy␣(MHz)')
```

```
    plt.legend()
    plt.grid(True, alpha=0.3)
    plt.show()
    return B_values, energies

def time_varying_strain_simulation(self, t_max=10, n_steps=2000,
                                   B_parallel=0, B_perpendicular=0,
                                   sigma_parallel=0, A_iso=0,
                                   strain_freq=1, strain_amp=1e6):
    """
    Simulate the NV center energy levels as a function of time-
        varying strain.
    sigma_x and sigma_y vary sinusoidally with time.
    """
    t_values = np.linspace(0, t_max, n_steps)
    # Define time-dependent strain components
    sigma_x_values = strain_amp * np.sin(2 * np.pi * strain_freq *
        t_values)
    sigma_y_values = strain_amp * np.cos(2 * np.pi * strain_freq *
        t_values)

    energies_vs_time = []

    for i, t in enumerate(t_values):
        # Print matrix at beginning, middle, and end for debugging
        print_matrix = (i == 0) or (i == len(t_values) // 2) or (i
            == len(t_values) - 1)
        H = self.build_hamiltonian(B_parallel, B_perpendicular,
                                   sigma_parallel,
                                   sigma_x_values[i], sigma_y_values
                                       [i],
                                   A_iso,
                                   print_matrix=print_matrix)
        evals, _ = self.diagonalize(H)
        energies_vs_time.append(evals)

    energies_vs_time = np.array(energies_vs_time).T  # Shape: (3,
        n_steps)
```

```
    # Plotting: energy levels
    fig = plt.figure(figsize=(15, 10))
    gs = GridSpec(2, 1, height_ratios=[2, 1])
    ax1 = fig.add_subplot(gs[0])
    spin_labels = ['$m_s=0$', '$m_s=+1$', '$m_s=-1$']
    for i in range(3):
        ax1.plot(t_values, energies_vs_time[i]/1e6, label=
            spin_labels[i], linewidth=2)
    ax1.set_title('Energy Levels Under Time-Varying Strain')
    ax1.set_ylim(-3000, 10000)
    ax1.set_ylabel('Energy (MHz)')
    ax1.grid(True, alpha=0.3)
    ax1.legend()

    # Plotting: strain signals
    ax2 = fig.add_subplot(gs[1])
    ax2.plot(t_values, sigma_x_values / 1e6, label=r'$\sigma_x(t)$ [
        MHz]', color='red')
    ax2.plot(t_values, sigma_y_values / 1e6, label=r'$\sigma_y(t)$ [
        MHz]', color='blue')
    ax2.set_xlabel('Time (arb. units)')
    ax2.set_ylabel('Strain (MHz)')
    ax2.grid(True, alpha=0.3)
    ax2.legend()
    plt.tight_layout()
    plt.show()

    return t_values, energies_vs_time, sigma_x_values,
        sigma_y_values

@staticmethod
def find_anti_crossings(B_values, energies, threshold=1e7):
    """
    Identify anti-crossing points where the gap between energy
        levels
    reaches a local minimum below a defined threshold.
    """
    energies = np.atleast_2d(energies)
    if energies.shape[0] < energies.shape[1]:
```

```
    energies = energies.T  # Ensure rows = levels

anti_crossings = []
num_levels, num_points = energies.shape

# Loop over adjacent levels
for i in range(num_levels - 1):
    # Compute gap between adjacent levels
    gaps = np.abs(energies[i + 1] - energies[i])
    minima = argrelextrema(gaps, np.less)[0]  # Local minima
        indices

    for idx in minima:
        if gaps[idx] < threshold:
            anti_crossings.append((B_values[idx], gaps[idx], i,
                i + 1))

print(f"\nFound␣{len(anti_crossings)}␣anti-crossing(s):")
for B, gap, i1, i2 in anti_crossings:
    print(f"Between␣levels␣{i1}␣and␣{i2}␣at␣B␣=␣{B:.2f}␣G␣with␣
        gap␣=␣{gap␣/␣1e6:.3f}␣MHz")
return anti_crossings
def time_dependent_population(self, initial_state_index=1,
                              B_parallel=0, B_perpendicular=0,
                              sigma_parallel=0, A_iso=0,
                              strain_freq=1e6, strain_amp=1e6,
                              t_max=5e-6, n_steps=2000):
"""
Simulates the NV center's quantum state evolution under time-
    varying strain.
Solves the time-dependent Schr dinger equation using `solve_ivp
    `.

Parameters:
- initial_state_index: index of the eigenstate to start in (0,
    1, or 2)
- strain_freq: frequency of oscillating strain (Hz)
- strain_amp: amplitude of strain oscillation (Hz)
- t_max: total simulation time (seconds)
```

```
- n_steps: number of time steps in the simulation
"""

# Time vector
t_values = np.linspace(0, t_max, n_steps)

def hamiltonian(t):
    # Strain components vary sinusoidally with time
    sigma_x = strain_amp * np.sin(2 * np.pi * strain_freq * t)
    sigma_y = strain_amp * np.cos(2 * np.pi * strain_freq * t)
    # Construct instantaneous Hamiltonian
    return self.build_hamiltonian(B_parallel, B_perpendicular,
                                  sigma_parallel, sigma_x,
                                     sigma_y, A_iso)

def schrodinger_rhs(t, psi):
    """
    Right-hand side of the time-dependent Schr dinger equation:
     d /dt = -i/   * H(t) *
    Using    = 1 (natural units), so:
     d /dt = -i * H(t) *
    """
    H_t = hamiltonian(t)
    return -1j * H_t.dot(psi)

# Initialize in one of the eigenstates at t=0
H0 = hamiltonian(0)
_, evecs = self.diagonalize(H0)
psi0 = evecs[:, initial_state_index]

# Solve the time-dependent Schr dinger equation
solution = solve_ivp(schrodinger_rhs, (0, t_max), psi0, t_eval=
   t_values, rtol=1e-8)

# Compute state populations over time for each basis state
populations = np.abs(solution.y) ** 2  # shape: (3, n_steps)

# Plot population dynamics
plt.figure(figsize=(12, 6))
```

```
for i in range(3):
    plt.plot(t_values * 1e6, populations[i], label=f'State␣{i}',
        linewidth=2)
plt.title('Quantum␣State␣Population␣Evolution␣Under␣Phonon␣
   Driving')
plt.xlabel('Time␣(  s)')
plt.ylabel('Population␣Probability')
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()

return t_values, populations, solution
```